Certified Local Optimality in a Constrained Landing Model

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A smooth airborne trajectory must satisfy force and control-rate limits at every time, clear an assumed occupied geometry, and connect to a feasible touchdown and stopping motion. We study this problem in a twelve-parameter polynomial family with a fixed hybrid contact and braking law. For a declared constant wind, we prove existence of an exact active solution with primary normalized margin near 0.1523607643018617. An outward-rounded Krawczyk calculation isolates a unique root of nine equations, coupling five flare variables, the objective value and three interior event times. A validated positive simplex of six active gradients supplies a strict first-order max-min sufficiency argument, rather than only necessary stationarity conditions. Stationary-event inequalities and a complete interval time cover establish feasibility of the connected landing. Dependence of the active margins on the flare variables alone transfers the result to non-strict local primary optimality in the full family. Analytical stopping and coupled energy–thrust inequalities give a separate, unmatched global-family upper bound of 0.181970215. We also establish continuous primary retention for a delivered secondary trajectory and certified feasibility of small known-wind parameter domains. The conclusions concern the explicit reduced model and conservative assumed geometry; global primary optimality, secondary-effort optimality and an operational feedback landing policy remain open.

Version 1 · 14 October 2026

Independent mathematical research note. Companion to *Checkers with Runway 13*, Version 5.

1. Introduction

A landing is a constrained connection between two different dynamical regimes. During flight, the shape and timing of the trajectory determine lift, thrust, bank and their rates. At touchdown, geometric contact replaces airborne motion; stopping then consumes the remaining runway. A trajectory can respect each endpoint requirement yet fail between endpoints, and a feasible numerical construction need not be locally optimal. The mathematical problem is to establish both statements for the same precisely defined object.

This note considers the assumption-conditional model A1 developed alongside *Checkers with Runway 13* (Ayyar 2026a,b). Its variables describe a seventh-degree approach joined to a sixth-degree flare, followed by ideal contact, a fixed delay and constant-deceleration ground roll. The primary objective is the smallest normalized operating margin over the complete trajectory. Geometry is imposed as a hard constraint. The optimization is finite-dimensional, but its inequality system is indexed by continuous time.

The principal result concerns five flare parameters. Six limiting margins meet at a common value: stopping, three thrust-slew events, maximum load and minimum thrust. The interior event times are unknowns, so equal margins and stationarity define nine equations. A verified root is only the first step. We prove that the active gradients form a simplex containing zero in its strict interior. Consequently every nonzero small flare perturbation decreases at least one necessary margin to first order. A uniform Taylor estimate makes this a strict local sufficiency theorem. Continuous-time verification then connects this upper-support argument to an actually feasible landing.

The approach has seven additional parameters. The six necessary active margins do not depend on them: the gate state is fixed and their evaluation is entirely within the flare or ground roll. This separation proves a non-strict local optimum of the twelve-parameter family. It neither identifies a unique approach nor permits arbitrary variations outside that family. An independent family-wide exclusion gives the bracket

$$0.152360764301861701 90 \le \eta_{\mathcal{P}}^{*} \le 0.181970215. \tag{1}$$

The left side is constructive; the right side is a necessary bound derived after discarding constraints. Their positive separation is the remaining global question.

The proof combines explicit analysis with finite interval computations. We give the contraction and local-sufficiency arguments in full, state exactly what the interval calculations enclose, and specify the immutable numerical inputs. Krawczyk's inclusion method (Krawczyk 1969) and verified linear algebra (Rump 2010) are used as mathematical tools, not as substitutes for the argument connecting their outputs to optimality. Decimal display values, numerical trajectory plots and forward replay have different evidential roles from the exact root box and outward inequalities.

The model is a declared mathematical scenario. It uses source-motivated aircraft dimensions together with assumed force limits, a conservative occupied set and an ideal contact law. The result does not certify a historical flight. The earlier synthetic model in V5 is retained as a separate mathematical object; its coefficients and certified values are not substituted into A1.

2. Mathematical model and optimization problem

2.1 Coordinates, state and force inversion

Lengths, times and forces are in metres, seconds and newtons unless indicated otherwise. The runway frame has x forward from the threshold, y to the right and h vertically above runway level. Wind $w=(w_x,w_y,0)$ is a velocity of air relative to ground; positive w_x is a tailwind. The baseline is the exact decimal vector

$$w^0=(-4.876589552990943,\ 9.64006667211232,\ 0). \tag{2}$$

The decision reference $r=(x,y,h)$ is virtual. The centre of gravity (CG) is

$$R=r+6e_z, \qquad \dot{R}=\dot{r}, \quad \ddot{R}=\ddot{r}. \tag{3}$$

The offset is fixed in the inertial vertical direction, not in the rotating body. Thus r is not a wheel point during bank. We use $z=R_z=h+6$ throughout all aerodynamic expressions.

Let $v=\dot{R}-w$, $V=|v|$ and $H=(v_x^2+v_y^2)^{1/2}$. On the regular domain $V,H>0$, define the orthonormal basis

$$e_t=v/V, \qquad e_r=(-v_y,v_x,0)/H, \qquad e_u=e_t\times e_r. \tag{4}$$

For $F=m(\ddot{R}+ge_z)$, set $L_u=F\cdot e_u$, $L_r=F\cdot e_r$, $L=(L_u^2+L_r^2)^{1/2}$ and $\phi=\operatorname{atan2}(L_r,L_u)$. Lift points along $\cos\phi\, e_u+\sin\phi\, e_r$. The reduced Newton equation is

$$m\ddot{R}=(T-D)e_t+L\cos\phi\, e_u+L\sin\phi\, e_r-mge_z. \tag{5}$$

Writing $Q=\rho(z)V^2S/2$, inverse dynamics gives

$$\rho(z)=1.225e^{-z/8500}, \qquad C_L=L/Q, \qquad n=L/(mg),$$

$$D=Q(0.08+0.045C_L^2), \qquad T=D+F\cdot e_t. \tag{6}$$

Here $\rho_0=1.225$, $m=238461.67357166$, $g=9.80665$ and $S=541.2$ are fixed model constants. Equivalent airspeed in knots and air flight-path angle are

$$E=\frac{Ve^{-z/17000}}{k}, \qquad \gamma=\operatorname{atan2}(v_z,H), \qquad k=1852/3600. \tag{7}$$

Bank and thrust rates are exact one-sided time derivatives within each flight phase. In particular,

$$\dot{\phi}=(L_u\dot{L}_r-L_r\dot{L}_u)/L^2. \tag{8}$$

The moving basis is differentiated as well as the acceleration; the full formulas are in Appendix B.

2.2 The twelve-parameter family

Let

$$q=(T_a,T_f,x_D,U,b_x,c_x,b_y,c_y,b_z,c_z,f_x,f_z)\in\mathcal{B}.$$

The closed parameter box is given in Table 1. Each phase uses normalized time $s\in[0,1]$; its physical derivatives are $T^{-j}d^j/ds^j$.

Table 1
The unchanged polynomial parameter domain. Bump amplitudes are in metres.

Coordinates	Closed interval
T_a,T_f	$[30,55]$, $[4,12]$ seconds
x_D,U	$[200,700]$ metres, $[69,88]$ metres/second
b_x,c_x,b_y,c_y	$[-400,400]$ each
b_z,c_z,f_x,f_z	$[-50,50],[-50,50],[-50,50],[-10,10]$

For endpoint positions P_0,P_1, velocities v_0,v_1 and zero accelerations, the quintic Hermite polynomial is

$$\begin{aligned}\mathcal{H}_T(s)&=P_0+Tv_0s+(10A-4B)s^3\\&\quad+(-15A+7B)s^4+(6A-3B)s^5,\\A&=P_1-P_0-Tv_0,\qquad B=T(v_1-v_0).\end{aligned}\tag{9}$$

Put $b(s)=64s^3(1-s)^3$ and $o(s)=(2s-1)b(s)$. The approach adds $(b_xb+c_xo,b_yb+c_yo,b_zb+c_zo)$; the flare adds $(f_xb,0,f_zb)$. Both additions and their first two derivatives vanish at the endpoints.

The CG positions are $P_A=(-2500,1000,146)$, $P_G=(0,0,21.24)$ and $P_D=(x_D,0,6)$. At entry and gate, $E=153$, $\gamma=-3.1^\circ$, with ground-track directions -47° and 0°. For unit horizontal ground direction $d=(d_x,d_y,0)$, the endpoint velocity is

$$\begin{aligned}V_z&=153ke^{z/17000},\\G_d&=w\cdot d+\sqrt{(V_z\cos\gamma)^2-|w|^2+(w\cdot d)^2},\\\dot R&=(G_dd_x,G_dd_y,V_z\sin\gamma).\end{aligned}\tag{10}$$

The positive square-root branch and $G_d>0$ are part of the endpoint specification. At touchdown, $\dot R=(U,0,-\sigma)$ with $\sigma=0.7$. Both phase endpoint accelerations are zero. Position, velocity and acceleration therefore join exactly at the gate. Control rates need not match; both one-sided values are constrained. The flare has $y\equiv0$.

The inherited corridor is also an exact part of the definition. Let $\mathcal{R}$ be the 2,001 CG-reference points used by the sealed checker, with their stored binary64 coordinate values interpreted exactly. The frozen

`reference_cloud.npy` already stores the CG-shifted cloud; the checker independently reconstructs the upward shift and verifies equality to this finite object. Its provenance is the original zero-wind synthetic reference with parameters $(36,6.5,500,77,0,\ldots,0)$ sampled at 1,001 equally spaced normalized times per phase (the common gate once). Those provenance formulas do not replace the stored coordinates by ideal real-arithmetic samples. The corridor margin is $1-\mathrm{dist}(R,\mathcal{R})/400$. This is distance to a finite point set, not to a fitted continuous centreline; time is continuous in its subsequent evaluation.

2.3 Operating margins, hard constraints and objective

For a two-sided operating quantity $f\in[\ell,u]$, define margins $(f-\ell)/(u-\ell)$ and $(u-f)/(u-\ell)$. For $|f|\le c$, define $1-|f|/c$. Table 2 specifies all airborne operating inequalities. They hold on both closed phases unless indicated otherwise.

Table 2
Normalized operating constraints. These, and the terminal margins below, determine the objective.

Quantity	Bounds or cap	Normalization
E	$[135,175]$ kt	span 40 kt
C_L	$[0,2.8]$	span 2.8
n	$[0.5,1.4]$	span 0.9
T	$[0,700000]$ N	span 700000 N
ϕ	$\lvert\phi\rvert\le 25°$	cap 25°
$\dot\phi$	$\lvert\dot\phi\rvert\le 5°/\mathrm{s}$	cap 5°/s
$\dot T$	$\lvert\dot T\rvert\le 100000$ N/s	cap 100000 N/s
γ	$\gamma\ge -6°$	$1+\gamma/(6°)$
Approach $\dot z$	$\dot z\ge -8$ m/s	$1+\dot z/8$
Corridor	distance ≤ 400 m	$1-\mathrm{dist}/400$

The touchdown distance margins are $(x_D-200)/500$ and $(700-x_D)/500$. The sink margins are $\sigma/1.5$ and $(1.5-\sigma)/1.5$. Lateral position, track and bank at touchdown are structurally zero, hence their normalized tolerance margins are identically one. The stopping margin is defined below. Forward ground velocity $\dot x\ge 0$, upward lift component $L_u>0$, all occupied-geometry predicates and intended gear contact are hard constraints, not additional objective margins.

For a geometrically admissible regular path define

$$\eta(q)=\min\left\{\inf_{s\in[0,1]} m_{j,\nu}(q,s),\ m_{k,D}(q),\ m_{\mathrm{stop}}(q)\right\}, \tag{11}$$

where $\nu\in\{a,f\}$ and the indices range over precisely the applicable margins above. Set $\mathcal{P}=\{q\in\mathcal{B}:\text{hard constraints hold},\ \eta(q)\ge 0\}$ and $\eta^*_{\mathcal{P}}=\sup_{q\in\mathcal{P}}\eta(q)$.

The secondary functional is

$$\mathcal{E}(q)=\sum_{\nu=a,f}\int_0^{T_\nu}\left[\left(\frac{\phi}{25^\circ}\right)^2+\left(\frac{\dot\phi}{5^\circ/\mathrm{s}}\right)^2+\left(\frac{\dot T}{100000}\right)^2\right]dt. \tag{12}$$

It has units of time. The intended secondary problem minimizes it subject to retaining a primary target within 10^{-4}. A numerical minimizer and a certified retention inequality are distinct conclusions.

2.4 Assumed occupancy and obstacle geometry

The CG-fixed non-gear box is $[-40.8,30.2]\times[-32.7,32.7]\times[-4,13.8]$. Body pitch is zero, yaw is ground track and body bank equals ϕ. The ideal gear layout has vertical coordinates -6 and maximum absolute lateral coordinate 6.0579. In the allowed bank range the lowest body and gear heights are

$$z_B=z-4\cos\phi-32.7|\sin\phi|,\quad z_G=z-6\cos\phi-6.0579|\sin\phi|. \tag{13}$$

All occupied points fit in a horizontal disk of radius 55; for example the largest body-corner norm is less than 55. With horizontal allowance $b_h=5$, write $r_h=55+b_h=60$ and $z_-=\min(z_B,z_G)$.

For a rectangle $[\ell_x,u_x]\times[\ell_y,u_y]$ with assumed top h_o above runway, impose

$$\max\left\{\max(\ell_x-x,x-u_x,\ell_y-y,y-u_y)-r_h,z_--h_o-3\right\}\ge 0. \tag{14}$$

This sufficient separation test defines the conservative feasible set. Its failure is not a collision theorem for a more detailed shape. In addition, $z_B\ge 0.5$ and $z_G\ge 0$ hold throughout flight.

The complete assumed scene consists of the frozen source-derived bounding rectangles, an assumed hill cap and light-point rectangles, together with an urban blanket of height 25 above runway. The blanket covers $x\le -600$ and the side regions $-600\le x\le 0$, $|y|\ge 100$, using the same inflated-rectangle disjunction. Source structures receive assumed tops of 45 m HKPD; hill and light tops are 140 and 12 m HKPD. Runway level is assumed 6 m HKPD. The adopted threshold easting/northing is (838277.9926795285,820320.961097593) m and grid bearing 134.02°. These choices define a frame and scene, not a datum or historical registration result. The exact rectangles are a finite input in `scene.json`, bound by the source manifest.

2.5 Ideal contact and ground roll

At $t_D=T_a+T_f$, bank is zero and all ideal gear points reach flat ground together. The model resets vertical velocity from $-\sigma$ to zero and pose to straight, level ground motion. Its translational vertical impulse is $m\sigma$, with kinetic-energy removal $m\sigma^2/2$; no strut-load or attitude-moment dynamics are asserted. Aerodynamic control-rate constraints end at touchdown and do not apply to the ideal reset. Lateral tyre authority and successful braking mode are assumptions.

With $u = t - t_D$, delay $\tau = 2$ and deceleration $a_b = 1.4709975$, the ground law is

$$x(u) = \begin{cases} x_D + Uu, & 0 \le u \le \tau, \\ x_D + U\tau + U(u-\tau) - \frac{1}{2}a_b(u-\tau)^2, & \tau \le u \le \tau + U/a_b, \\ x_D + \tau U + U^2/(2a_b), & u \ge \tau + U/a_b. \end{cases} \tag{15}$$

Throughout this phase $y = 0, z = 6$. Let $L_0 = 2786 - 150 = 2636$. The exact stopping margin is

$$m_{\text{stop}} = \frac{L_0 - x_D - \tau U - U^2/(2a_b)}{L_0}. \tag{16}$$

Hard obstacle separation is checked along the entire ground segment. Contact with the runway is a structural equality, not a clearance tolerance.

2.6 Relation to the synthetic V5 model

V5's original family uses the unshifted point reference at heights 140, 15.24 and zero and evaluates density there. A1 uses CG heights 146, 21.24 and 6 in both endpoint speed construction and density. Although (3) preserves acceleration, those altitude-dependent speeds change polynomial coefficients and forces. The two models therefore cannot be interchanged by translating a plotted path alone. A1 also replaces the empty synthetic obstacle scene with the declared whole-aircraft geometry and ideal contact interpretation above. The inherited parameter box, force constants, operating normalizations and braking constants remain unchanged.

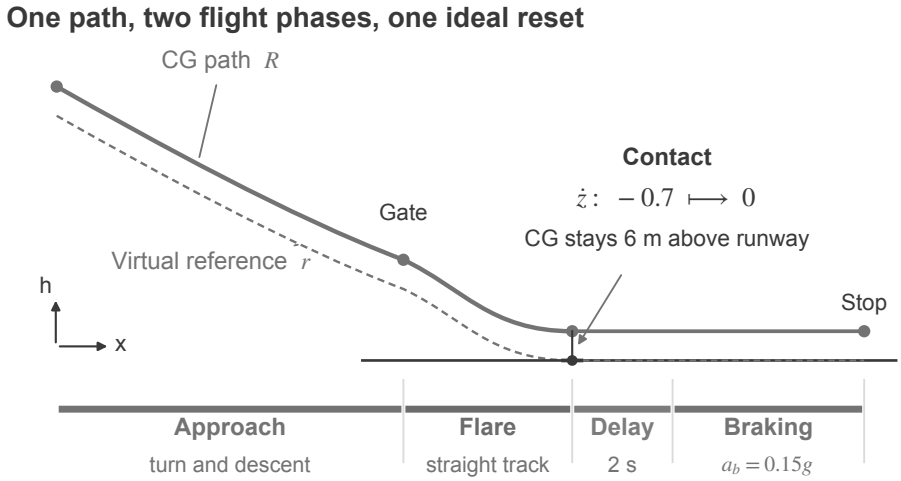


Figure 1
The hybrid landing object, shown schematically. The virtual reference and CG differ by a fixed vertical translation during flight; the gear is part of a banked occupied set. Contact resets the vertical motion before delay and braking. Heights and phase lengths in this schematic are not to scale.

3. Analytical bounds and preliminary results

3.1 Stopping and terminal restrictions

Lemma 1 (Stopping monotonicity). For $a_b > 0$, $\tau \geq 0$ and $U \geq 0$, the stopping position $S(x,U) = x + \tau U + U^2/(2a_b)$ is increasing in x and strictly increasing in U on $(0,\infty)$. At required margin e, a nonnegative speed satisfies $S(x,U) \leq L_0(1-e)$ if and only if $B = L_0(1-e) - x \geq 0$ and

$$0 \leq U \leq -a_b\tau + \sqrt{a_b^2\tau^2 + 2a_bB}. \tag{17}$$

Proof. $S_x = 1$ and $S_U = \tau + U/a_b > 0$ for $U > 0$. Multiplying the stopping inequality by $2a_b$ and completing the square gives $(U + a_b\tau)^2 \leq a_b^2\tau^2 + 2a_bB$. The right endpoint in (17) is nonnegative exactly when $B \geq 0$. The ground equations integrate to S, so no stopping approximation is involved. ∎

At touchdown $V_D^2 = (U - w_x)^2 + w_y^2 + \sigma^2$. The lower EAS margin gives $V_D \geq k(135 + 40e)\exp(6/17000)$. Set

$$D_e = [k(135 + 40e)\exp(6/17000)]^2 - w_y^2 - \sigma^2, \qquad U_e = w_x + \sqrt{D_e}. \tag{18}$$

We explicitly restrict the following terminal relaxation to $D_e > 0$ and the positive air-forward touchdown branch $U - w_x > 0$, with $U_e > 0$. For baseline headwind this branch already follows from $U > 0$; it must not be assumed merely from positive ground speed for arbitrary tailwind.

Proposition 1 (Terminal relaxation). On $0 \le e \le 0.45$ and the branch above, any feasible landing satisfying the prescribed terminal conditions obeys

$$f(e) = 200 + 500e + \tau U_e + \frac{U_e^2}{2a_b} - L_0(1-e) \le 0. \tag{19}$$

The function f is strictly increasing. If $f(0) > 0$, no nonnegative-margin landing exists on that branch. If $f(0) \le 0 < f(0.45)$, its unique zero in $[0,0.45)$ is the relaxed upper value; if $f(0.45) \le 0$, the relaxed upper value is 0.45. This relaxation need not produce an airborne trajectory.

Proof. The touchdown-distance margin requires $x_D \ge 200 + 500e$, while (18) and the branch choice require $U \ge U_e$. Lemma 1 gives (19). The entry EAS is 153, hence its lower normalized margin is $(153-135)/40 = 0.45$. Differentiation yields

$$U_e' = \frac{40k^2 \exp(12/17000)(135+40e)}{\sqrt{D_e}} > 0,$$

$$f'(e) = 3136 + (\tau + U_e/a_b)U_e' > 0.$$

Continuity and strict monotonicity prove the alternatives, including equality at $e = 0$. Conversely, whenever $f(e) \le 0$, choosing $x_D = 200 + 500e$ and $U = U_e$ satisfies these retained inequalities. For $e \le 0.45$, the touchdown upper-distance and fixed-sink margins do not obstruct this relaxed choice. Other operating, geometric and path conditions have been removed; they cannot be restored by the scalar argument. ∎

The zero has an algebraic representation. With

$$Q_e = 200 - L_0 + 3136e + \tau w_x + \frac{w_x^2 + D_e}{2a_b}, \qquad B_w = \tau + w_x/a_b,$$

it solves $Q_e + B_w\sqrt{D_e} = 0$, hence $Q_e^2 - B_w^2 D_e = 0$, a quartic in e. Squaring is not reversible without the original sign equation. In particular B_w can be negative; if $B_w = 0$, the unsquared equation is simply $Q_e = 0$.

The polynomial family additionally imposes $U \ge 69$. Replacing U_e by $\max(69, U_e)$ preserves the bound. When the floor binds at the relaxed optimum, its exact value is

$$e_{69} = \frac{L_0 - 200 - 2(69) - 69^2/(2a_b)}{3136} = \frac{66656817}{307536544} \approx 0.21674437819. \tag{20}$$

The switch satisfies $D_e = (69 - w_x)^2$ with $69 - w_x \ge 0$. The broader baseline terminal relaxation has upper value approximately 0.223501289399. Neither value is the achievable family optimum; the much smaller bound below uses airborne work.

3.2 Regularity and attainment

All polynomial times lie in compact intervals bounded away from zero. The exact Bernstein calculation in the source regularity certificate gives

$$|\dot{z}_a| \le \frac{33792275107}{1258291200} < 26.856, \qquad |\dot{z}_f| \le \frac{719106075}{33554432} < 21.432. \tag{21}$$

These bounds hold over the entire parameter box, not only along the delivered path. They use the bound 5 on the magnitudes of the fixed vertical endpoint velocities. Appendix A explains the finite rational calculation.

Lemma 2 (Uniform force regularity). On any nonnegative-margin portion of the declared family satisfying the hard ground and bank constraints, V, H, L, L_u are uniformly separated from zero. Thus inverse forces and phasewise control rates extend continuously to its closure.

Proof. For $|\phi| \le 25°$, gear nonpenetration gives $z \ge 6\cos\phi + 6.0579|\sin\phi| \ge 6$. The last inequality follows by concavity on $[0, 25°]$: the smaller endpoint value is 6. Hence $V \ge 135k\exp(6/17000) > 69$. With zero vertical wind, (21) gives $H^2 = V^2 - \dot{z}^2$ bounded strictly away from zero. Also $L \ge 0.5mg$ and $L_u = L\cos\phi \ge 0.5mg\cos 25° > 0$. All denominators in the force and rate formulas are consequently bounded away from zero. Polynomial dependence and the smooth elementary operations then give continuity, including at phase endpoints when each side is retained separately. ∎

Proposition 2 (Attainment on the closed admissible family). For the declared closed hard predicate $\dot{x} \ge 0$, a nonnegative feasible witness implies that the maximum primary value is attained. For any nonempty closed primary-retention superlevel, the secondary effort minimum is attained.

Proof. Consider a sequence in the closed admissible family. Compactness of $\mathcal{B}$ gives a convergent subsequence. Lemma 2 supplies uniform positive denominator bounds along the sequence and its limit, so every force and rate quantity converges uniformly on each normalized phase. The finitely many geometric maxima and minima are continuous; inequalities for every time are preserved under this convergence. Endpoint equalities and the ground law are continuous in q. Thus the admissible set is closed in a compact box. For finitely many continuous margin functions on compact normalized-time domains, $|\inf m(q,\cdot) - \inf m(q',\cdot)| \le \sup|m(q,\cdot) - m(q',\cdot)|$, which proves continuity of η. The extreme-value theorem applies to η and, on a closed retention set, to the continuous integral $\mathcal{E}$. The computed witness has strictly positive forward speed, a stronger property than the closed model constraint. ∎

3.3 Coupled airspeed, energy and thrust-slew restrictions

For this subsection $e\in[0,0.45]$ is a tested margin. The flare has $\dot y=0$. The family bounds imply $6\le z\le 34$ and $|\dot z|\le 22$; these constants are not available for an arbitrary smooth path. Define

$$v_-=k(135+40e),\qquad v_+=k(175-40e),$$

$$V_-=v_-\exp(6/17000),\qquad V_+=v_+\exp(34/17000),$$

$$Q_-=\tfrac12\rho_0 S v_-^2,\qquad Q_+=\tfrac12\rho_0 S v_+^2,$$

$$L_+=mg(1.4-0.9e),$$

$$D_+=\max_{Q\in\{Q_-,Q_+\}}\left(0.08Q+\frac{0.045L_+^2}{Q}\right). \tag{22}$$

The last expression bounds drag: $Q=\rho_0 S(kE)^2/2$ exactly, drag increases with L^2, and its displayed dependence on $Q>0$ is convex. Dropping the simultaneous C_L constraint only enlarges this upper bound.

Assume $V_-^2-w_y^2-22^2>0$. At the gate $v_x>0$; since $v_x^2\ge V_-^2-w_y^2-22^2$, continuity prevents a sign change. Consequently the forward air branch is valid for the entire flare and

$$\dot x\ge u_-=w_x+\sqrt{V_-^2-w_y^2-22^2}>0. \tag{23}$$

The final strict inequality is checked for each reported condition. For $u=\dot x$,

$$(V/u)^2\le 1-2w_x/u+(w_x^2+w_y^2+22^2)/u^2.$$

Its derivative has no positive local maximum: for $w_x\le 0$ it is negative; for $w_x>0$ the sole critical point is a minimum. Thus

$$R_v=\max\left(1,\frac{\sqrt{(u_--w_x)^2+w_y^2+22^2}}{u_-}\right),\qquad \int_0^{T_f}V\,dt\le R_v x_D. \tag{24}$$

At the zero-acceleration gate, writing $\theta=3.1^\circ$,

$$Q_G=\tfrac12\rho_0 S(153k)^2,\quad L_G=mg\cos\theta,$$

$$T_G=-mg\sin\theta+0.08Q_G+0.045L_G^2/Q_G. \tag{25}$$

The lower thrust margin and slew cap imply $T(t)\ge 700000e$ and $T(t)\ge T_G-Kt$, $K=100000(1-e)$. Put

$$A=\max(0,T_G-700000e),\quad t_-(x)=\max(4,x/(w_x+V_+)),$$

$$z_*(x)=\min(t_-(x),A/K),\quad J(x)=Az_*(x)-Kz_*(x)^2/2. \tag{26}$$

Here $w_x+V_+>0$ and $T_f\ge t_-(x_D)$ follow from the family duration floor and $\dot x\le w_x+V_+$. Hence $J(x_D)$ is a lower bound for the integrated thrust excess above $700000e$.

Proposition 3 (Coupled necessary work inequality). Under these premises every feasible flare at margin at least e satisfies

$$U \geq \max\left\{69, U_e, w_x + \sqrt{\max(0, R - Cx_D + B(x_D))}\right\}, \tag{27}$$

where $V_G = 153k\exp(21.24/17000)$ and

$$R = V_G^2 + 2g(15.24) - w_y^2 - \sigma^2,$$
$$C = 2\max(0, D_+ - 700000e)R_v/m, \quad B(x) = 2V_- J(x)/m. \tag{28}$$

Proof. Dot (5) with v. Constant wind and $w_z = 0$ give the air-relative energy identity

$$\frac{d}{dt}\left(\frac{V^2}{2} + gz\right) = \frac{(T - D)V}{m}. \tag{29}$$

Since $T - D \geq -\max(0, D_+ - 700000e) + \max(A - Kt, 0)$, integration, (24) and $V \geq V_-$ yield

$$V_D^2 \geq V_G^2 + 2g(15.24) - Cx_D + B(x_D).$$

Substitute $V_D^2 = (U - w_x)^2 + w_y^2 + \sigma^2$. The positive air branch proved above allows the positive square root. If the resulting lower radicand is negative, $U > w_x$ still implies the weaker lower bound obtained by replacing it with zero. Combine with the terminal and family speed floors. ∎

Proposition 4 (Certified global-family exclusion). For the baseline wind and A1 model, no member of $\mathcal{P}$ has $\eta(q) \geq 0.181970215$. In particular the weak upper endpoint in (1) is valid.

Computer-assisted proof. Take the exact decimal $e = 0.181970215$. The possible touchdown interval is $[200 + 500e, 700 - 500e]$. Partition it into 4,096 equal closed cells with rational-decimal endpoints. On each cell use outward interval operations to enclose the lower-speed expression $\underline{U}(x)$ in (27) and then

$$x + 2\underline{U}(x) + \underline{U}(x)^2/(2a_b) - 2636(1 - e).$$

Every lower enclosure is strictly positive; the smallest displayed lower endpoint exceeds 0.9678323903 m. The code tests the unrounded interval endpoint against zero, not this displayed summary. Lemma 1 rules out stopping within the available length in every cell. Larger margins satisfy all original inequalities at e, so they are excluded by nesting; no monotonicity claim about the derived relaxation itself is needed. The source is `strong_bound.exclusion`, rerun without optimization in the accompanying verification package. ∎

The argument proves a necessary exclusion over the entire polynomial parameter family, not an optimal trajectory. In particular, (21), $z \le 34$ and $T_f \ge 4$ are family restrictions. The terminal proposition has a broader branch-qualified domain; the energy upper bound cannot be transferred to that unrestricted class. Earlier synthetic-model bounds and invalidated interval searches are separate records (Appendix G).

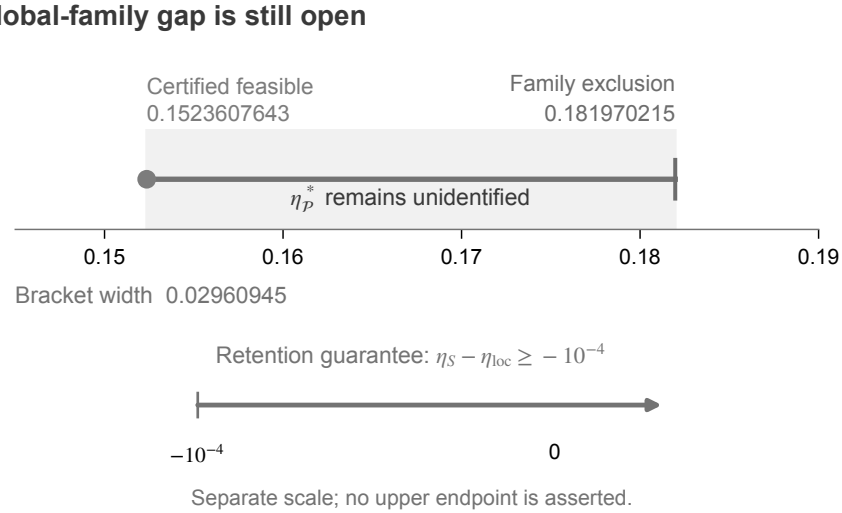


Figure 2
The proved feasible value and the family-wide exclusion bound are separated. The interval is a bracket for the unknown global family optimum, not an uncertainty interval for the local root. The secondary-retention scale is shown separately so that it cannot be confused with the much larger global gap.

4. Characterization of the active landing solution

Write $a = (T_a, b_x, c_x, b_y, c_y, b_z, c_z)$ for the approach coordinates and $p = (T_f, x_D, U, f_x, f_z)$ for the flare coordinates. Their feasible domains are the corresponding products in Table 1. The normalized coordinates used for root isolation are

$$\zeta = (T_f/5, x_D/300, U/75, f_x, f_z, \eta, s_S, s_L, s_T) \in \mathbb{R}^9. \tag{30}$$

Scales 5, 300 and 75 carry the respective physical units; all computations use their exact values. Let $\mathcal{T}(p,s)$ be flare thrust and $\dot{\mathcal{T}} = T_f^{-1}\partial_s \mathcal{T}$ its physical time

derivative. The six active margins are

$$\begin{aligned} g_1(p) &= m_{\mathrm{stop}}(p), \\ g_2(p) &= 1+\dot{\mathcal{T}}(p,0)/100000, \\ g_3(p) &= 1-\dot{\mathcal{T}}(p,1)/100000, \\ g_4(p,s_S) &= 1-\dot{\mathcal{T}}(p,s_S)/100000, \\ g_5(p,s_L) &= (1.4-n_f(p,s_L))/0.9, \\ g_6(p,s_T) &= \mathcal{T}(p,s_T)/700000. \end{aligned} \tag{31}$$

The event-time letters indicate interior slew, load and thrust. Gate slew here is the flare-side value; the approach-side value is checked separately.

The signed expressions agree locally with the absolute slew margins because gate slew is negative, while touchdown and interior slew are positive. The source derivation's sentence calling touchdown slew negative is a prose error: both executable equations and the sealed certificate record the sign vector $(-1,+1,+1)$. The new audit records this correction without changing the earlier artifact.

The nine equations are

$$F(\zeta) = (g_1-\eta,\dots,g_6-\eta,\partial_s g_4,\partial_s g_5,\partial_s g_6)^{\top} = 0. \tag{32}$$

Here the first five coordinates of ζ determine p, and the last three supply the corresponding interior events. Thrust slew is T_s/T_f; its stationarity is $T_{ss}/T_f = 0$. These identities determine the needed mixed derivatives exactly. At the certified solution the approximate physical values are

$$\begin{aligned} T_f &\simeq 4.140834462862, & x_D &\simeq 302.126459392735, \\ U &\simeq 72.512142316831, & f_x &\simeq -0.054163922082, \\ f_z &\simeq -0.014824733206. \end{aligned} \tag{33}$$

They summarize the solution but do not define it. The exact definition is the unique root in the box of Theorem 1, with the seven approach decimals specified in Appendix C.

A fixed-time upper-support argument will use the event times at the exact root. It is not necessary to optimize the event times anew for every perturbed p: a continuous-time minimum cannot exceed its value at any one fixed normalized time. This observation avoids differentiating an unknown global argmin map.

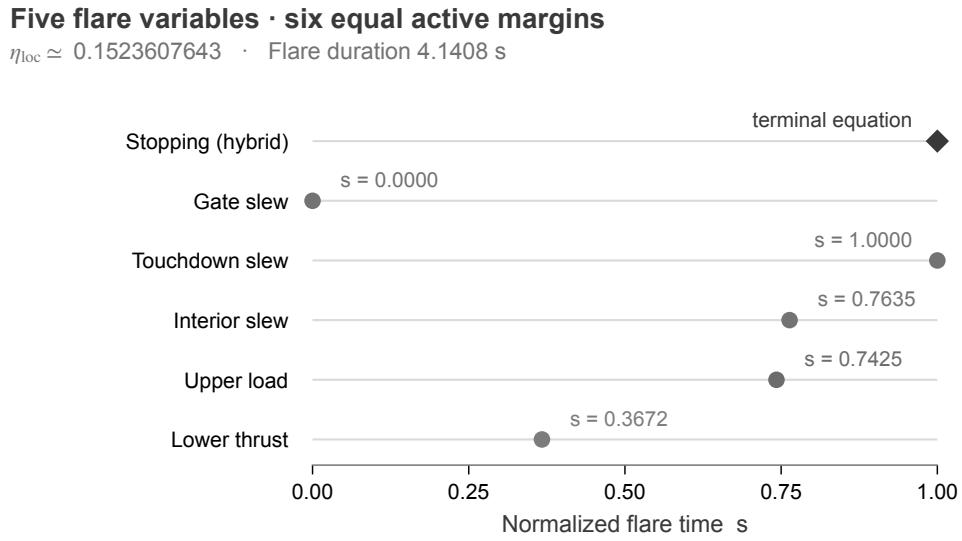


Figure 3
The six active margins and their locations in the flare. The three interior events are distinct; load and slew happen close together but are not the same stationary point. Event-time positions are evaluated from the numerical centre solely for illustration. Stopping is a terminal hybrid constraint, not an additional in-flight event.

5. Rigorous local-optimality theorem

5.1 Validated root and nonsingularity

Let $c \in \mathbb{Q}^9$ be the complete decimal centre printed in Appendix C, and put $r_0 = 10^{-20}$. The exact intended box is

$$X = c + [-r_0, r_0]^9. \tag{34}$$

Its scale is the coordinate system (30), not metres, seconds or a neighborhood on which optimality has been quantified. Outward interval conversion encloses this rational box by a slightly larger representable hull $\widehat{X}$.

Theorem 1 (Validated root existence and uniqueness). For the baseline wind (2) and the stated A1 force law, F in (32) has exactly one root ζ^* in X. Every Jacobian enclosed by the interval Jacobian calculation on $\widehat{X}$ is nonsingular. In particular

$$\begin{gathered} c_6 - 10^{-20} \le \eta_0 \le c_6 + 10^{-20}, \\ c_6 = 0.152360764301861701918625206455836893973748530913. \end{gathered} \tag{35}$$

where $\eta_0 = \zeta_6^*$. The three stationary-event coordinates lie strictly in $(0,1)$ and the signed slew branches are $(-1,+1,+1)$.

Computer-assisted proof. The force formulas are compositions of polynomials, exponentials, square roots and division with denominators separated

from zero on $\widehat{X}$. They therefore define a C^2 map on an open neighborhood of X. Interval automatic differentiation supplies an inclusion matrix $\mathbb{J}(\widehat{X})$ containing $DF(\zeta)$ for every $\zeta \in X$. Let C be the fixed matrix of exact finite decimal entries generated by rounding a high-precision inverse at c. Its entries are then re-entered as outward intervals; this approximate inverse is not assumed exact.

The verified Krawczyk enclosure is

$$K(X) = c - CF(c) + (I - C\mathbb{J}(\widehat{X}))(X - c). \tag{36}$$

The outward computation encloses every component of $K(X) - c$ in $(-1.964 \times 10^{-34}, 1.964 \times 10^{-34})$ and proves

$$\sup_{J \in \mathbb{J}(\widehat{X})} \|I - CJ\|_\infty < 1.964 \times 10^{-14} < 1. \tag{37}$$

The sharper component bounds accompany Table A1. These are comparisons of interval endpoints before decimal display rounding. All arithmetic used in the comparisons is outward-rounded at 50 decimal digits.

For any $\zeta \in X$, the integral mean-value formula gives

$$\zeta - CF(\zeta) = c - CF(c) + \left(I - C\int_0^1 DF(c + t(\zeta - c))\,dt\right)(\zeta - c) \in K(X).$$

Thus $\Psi(\zeta) = \zeta - CF(\zeta)$ maps X into its strict interior and, by (37), is a contraction in the infinity norm. The contraction theorem gives a unique fixed point in X. For any enclosed J, the matrix $CJ = I - (I - CJ)$ is invertible by its convergent Neumann series. In finite dimensions both C and J are consequently invertible. Hence fixed points of Ψ are precisely zeros of F. This proves existence, uniqueness in X and the nonsingularity assertion.

The interval inputs enclose the intended exact c and r_0; inclusion is checked against the inner radius, so use of $\widehat{X}$ does not weaken the conclusion to a different decimal box. The event-time and slew-sign assertions are direct separated interval inequalities. Newton iteration only selected c; neither convergence nor its residual supplies any of the existence premises. ∎

The proof establishes local root uniqueness inside the specified box. It says nothing about other roots outside it. Likewise nonsingularity at fixed baseline wind provides a local implicit branch, not a certified interval of wind values.

5.2 The active-gradient simplex

Fix (s_S, s_L, s_T) at their exact root values and write the six functions in (31) as $g_i(\xi)$ in the five scaled flare coordinates $\xi = (T_f/5, x_D/300, U/75, f_x, f_z)$. Let G have rows $\nabla g_i(\xi^*)^\top$ and define

$$A_G = \begin{pmatrix} G^\top \\ \mathbf{1}^\top \end{pmatrix}, \qquad A_G \lambda = (0,0,0,0,0,1)^\top. \tag{38}$$

The proof uses only these fixed-time functions; it does not differentiate an argmin map.

Lemma 3 (Validated positive simplex). The augmented matrix A_G is nonsingular, and its solution λ has strictly positive entries. In particular G has rank five, $G^\top\lambda = 0$, $\mathbf{1}^\top\lambda = 1$, and zero belongs to the strict interior of the simplex generated by the six gradient vectors.

Computer-assisted proof. Apply interval differentiation to the six margins over $\widehat{X}$, using only the first five derivatives in each row. For a fixed decimal approximate inverse C_A, the verified bound is $r_A = \|I - C_A A_G\|_\infty < 2.484 \times 10^{-14}$. The Neumann argument used above proves nonsingularity uniformly over the enclosed matrices. For a chosen centre $\widehat{\lambda}$ and $b = (0,0,0,0,0,1)^\top$, the residual equation gives

$$\left\|\lambda - \widehat{\lambda}\right\|_\infty \le \frac{\left\|C_A(b - A_G\widehat{\lambda})\right\|_\infty}{1 - r_A}. \tag{39}$$

The implementation encloses both centre and residual with intervals, so no unvalidated solve is used. The resulting outward summaries are in Table 3; every lower endpoint is positive. Rank five follows because the condition $\operatorname{rank}(G) < 5$ would make its five transposed rows, and hence the six rows of A_G, dependent. Nonsingularity of A_G also means affine independence of its six gradient columns. Positive barycentric coordinates summing to one put zero in the strict interior of that five-dimensional simplex. ∎

Table 3
Outward decimal enclosures of the simplex weights. The source stores longer endpoint strings; these deliberately widened intervals retain strict positivity.

Active margin	Lower endpoint	Upper endpoint
Stopping	0.8560644515753778	0.8560644515753782
Gate slew	0.0009658211277946	0.0009658211277950
Touchdown slew	0.0045094051150227	0.0045094051150231
Interior slew	0.0087917210360731	0.0087917210360735
Upper load	0.0514426236725329	0.0514426236725333
Lower thrust	0.0782259774731976	0.0782259774731981

Theorem 2 (Strict local primary optimality in five variables). Let p^* be the flare coordinates of ζ^*. Define the flare relaxation as the minimum of all continuous flare operating margins and the terminal margins, omitting the approach and hard geometric inequalities. Then p^* is a strict local maximum of this primary objective, with value η_0, once whole-time feasibility in Theorem 3 is established. There exist $\epsilon, \alpha > 0$ such that every nonzero scaled perturbation h with $\|h\| < \epsilon$ satisfies

$$\eta_f(\xi^* + h) \le \eta_0 - \alpha\|h\| < \eta_0. \tag{40}$$

No numerical value of ϵ or α is claimed.

Proof. For nonzero d, suppose $\nabla g_i(\xi^*) \cdot d \ge 0$ for all i. Multiplying by the strictly positive weights and summing gives zero. Each term must therefore vanish. Since G has rank five, $d = 0$, a contradiction. Thus every unit vector has a strictly negative minimum inner product.

The function $h_0(d) = \min_i \nabla g_i(\xi^*) \cdot d$ is continuous on the unit sphere. By compactness its maximum is a strictly negative number, say $-\beta$. Hence

$$\min_i \nabla g_i(\xi^*) \cdot h \le -\beta\|h\| \quad \text{for all } h.$$

Choose a closed ball around ξ^* in the regular signed-branch domain. The six Hessians are continuous and bounded there, so a common $M < \infty$ satisfies

$$|g_i(\xi^* + h) - g_i(\xi^*) - \nabla g_i(\xi^*) \cdot h| \le M\|h\|^2.$$

All six root values equal η_0. Taking their minimum gives

$$\min_i g_i(\xi^* + h) \le \eta_0 - \beta\|h\| + M\|h\|^2.$$

For $\|h\| < \beta/(2M)$ (with the evident interpretation if $M = 0$), this is at most $\eta_0 - \beta\|h\|/2$. Every full flare minimum is bounded above by these six fixed-time or terminal values. Theorem 3 establishes $\eta_f(\xi^*) = \eta_0$, completing (40) with $\alpha = \beta/2$ and a sufficiently small ϵ. ∎

This is a first-order sufficient max-min argument. Positive multipliers alone would not suffice without affine independence; necessary KKT conditions alone would not supply the uniform directional decrease. The fixed active-time construction also makes the proof independent of whether other local time extrema move smoothly under perturbation.

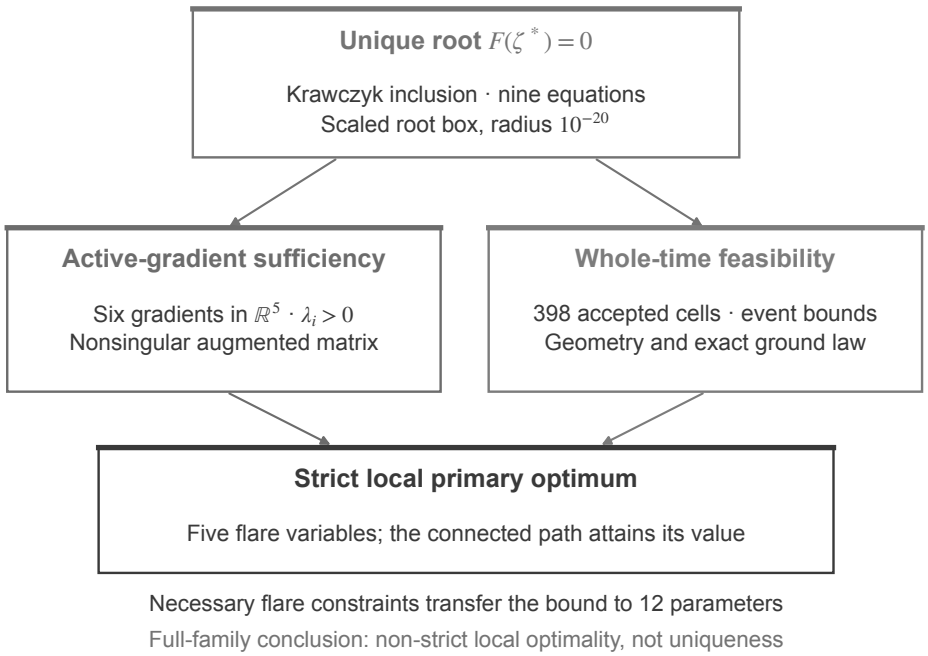


Figure 4
Logical structure of the certificate. The diagram depicts implications, not a projection of a five-dimensional simplex. Root isolation, active-gradient sufficiency and continuous-time feasibility supply different premises; only their combination supports the local-optimality conclusion.

6. Computer-assisted continuous-time verification

The local upper-support argument is useful only if the exact root path is feasible. Ordinary interval subdivision alone cannot usually prove an active equality: a cell containing a true minimum also contains values of the interval extension below it. The certificate therefore combines analytical local extrema with interval coverage of the remaining constraints.

6.1 Stationary neighborhoods and endpoint slopes

For this time-domain argument, extend the endpoint margins as $g_2(p,s) = 1+\dot{\mathcal{T}}(p,s)/100000$ near $s=0$ and $g_3(p,s) = 1-\dot{\mathcal{T}}(p,s)/100000$ near $s=1$. Their endpoint values are exactly the functions in (31). The five special time intervals are given in Table 4. Their endpoints denote the listed exact decimals; the computational intervals enclose those decimals outward. Each interior stationary coordinate of X lies strictly inside its assigned interval.

Lemma 4 (Local time-minimum bounds). At the exact root path, each special margin is at least η_0 throughout its corresponding interval in Table 4. Each of the three interior intervals contains exactly one stationary point of its assigned margin.

Table 4
Special normalized flare-time intervals and certified derivative signs. Derivatives are with respect to normalized time s.

Margin	Closed interval	Strict bound
Interior slew	[.7535208320375588, .773520832037559]	$g_4'' > 13.33$
Upper load	[.7325247126666857, .752524712666686]	$g_5'' > 6.58$
Lower thrust	[.3571596023161144, .37715960231611445]	$g_6'' > 0.869$
Gate slew	[0,.01]	$g_2' > 3.78$
Touchdown slew	[.99,1]	$g_3' < -2.31$

Computer-assisted proof. Interval Taylor jets enclose the required derivatives for every root parameter in $\widehat{X}$ and every time in the full assigned interval. They establish the strict curvature and slope bounds shown, while separating each slew sign. At the exact interior root time, the margin equals η_0 and its derivative is zero by (32). Strictly positive second derivative makes that stationary point the unique minimum on its interval. At the gate, a positive one-sided derivative implies the minimum is the endpoint value η_0; at touchdown the negative derivative gives the same conclusion from the right endpoint. These arguments require no assertion that all stationary points on the full phase have been enumerated. ∎

6.2 Coverage, omitted constraints and contact

For every other margin the checker uses an inclusion extension over closed time cells and the complete interval root box. It accepts a cell only if all ordinary operating margins have lower endpoints strictly greater than $c_6 + r_0$, hard geometry has nonnegative lower endpoints, and force regularity and forward speed hold. Inside a special interval it omits only the corresponding margin already controlled by Lemma 4. All other active and inactive margins remain tested there. Outside those intervals nothing is omitted.

A polynomial is evaluated after an exact shift to a cell centre, which reduces dependency overestimation. Repeated bisection refines unresolved cells; a depth limit leads to an unresolved result, never acceptance. The recorded accepted cells share their exact decimal endpoints, begin at zero and end at one separately in each phase. Hence their union covers every time, including both sides of the gate.

The corridor calculation selects one frozen reference point using a nearest-neighbour query, then encloses the distance to that point by interval arithmetic. For any selected point $P \in \mathcal{R}$, $\operatorname{dist}(R,\mathcal{R}) \le |R-P|$. The proof therefore does not trust the query to return the exact nearest point. A poor selection can

cause unnecessary subdivision, but cannot produce a false lower bound on the corridor margin.

At touchdown, direct interval subtraction cannot prove the exact zero gear clearance. The polynomial construction instead factors

$$z(s)-6=(1-s)h_1(s), \qquad \ddot{x}(s)=(1-s)a_1(s). \tag{41}$$

Since $\ddot{y}=0$ in the flare, $L_r=mw_y\ddot{x}/H$, so

$$z_G \geq (1-s)\left[h_1(s)-\frac{6.0579m|w_y a_1(s)|}{HL}\right]+6(1-\cos\phi). \tag{42}$$

The checker proves the bracket nonnegative; the last term is analytically nonnegative. Thus pre-contact nonpenetration includes the exact endpoint. This is a structural factorization, not a permitted penetration tolerance.

Theorem 3 (Whole-time feasibility of the connected landing). Retain the seven exact decimal approach coordinates in Appendix C and use the five exact flare coordinates of ζ^*. The resulting A1 path belongs to $\mathcal{P}$ and has continuous primary value $\eta(q^*)=\eta_0$. It satisfies all force, rate, corridor, hard occupied-geometry and hybrid stopping constraints, with the ideal contact transition specified in Section 2.5.

Computer-assisted proof. All twelve parameters lie strictly inside their original bounds. The interval calculation accepts 32 approach cells and 366 flare cells, for 398 accepted cells after 786 processed cells. Of these, 347 omit no margin; 45 omit only the already-proved upper-load margin, five only the already-proved slew margin, and one only the already-proved lower-thrust margin. The covers are closed and gap-free. The smallest ordinary checked operating slack over η_0 exceeds 6.86×10^{-7}; the interval acceptance comparisons use the larger upper root endpoint. Combining their inequalities with Lemma 4 proves every airborne operating margin is at least η_0.

The same cells establish object and urban separation, non-gear clearance and gear nonpenetration, including (42). Forward speed and L_u are strictly positive throughout this witness. Hermite and bump endpoint identities enforce the common gate state and zero accelerations, and impose touchdown lateral position, track and bank exactly. The touchdown distance and fixed-sink margins are strictly greater than the upper root endpoint. The stop margin equals η_0 by $F_1(\zeta^*)=0$.

For ground motion, substitute the interval-enclosed x_D, U into (15). Interval geometry is nonnegative over the entire segment $[x_D, x_D+2U+U^2/(2a_b)]$ at $y=0, z=6$. Ground gear contact is an exact structural equality. The stipulated reset and braking law complete the hybrid path; no airborne constraint is extended through the impulse. Since at least one margin equals η_0 and all others are no smaller, (11) has value exactly η_0. ∎

The rounded stop position is 2234.377025300292 m, leaving 551.622974699708 m of the full 2786 m runway. An independent forward integration of Newton's equation of the rounded numerical centre has maximum position discrepancy about 1.14×10^{-11} m. That number tests computational consistency of the force inversion; it neither gives physical accuracy nor replaces the interval proof for the exact implicit root.

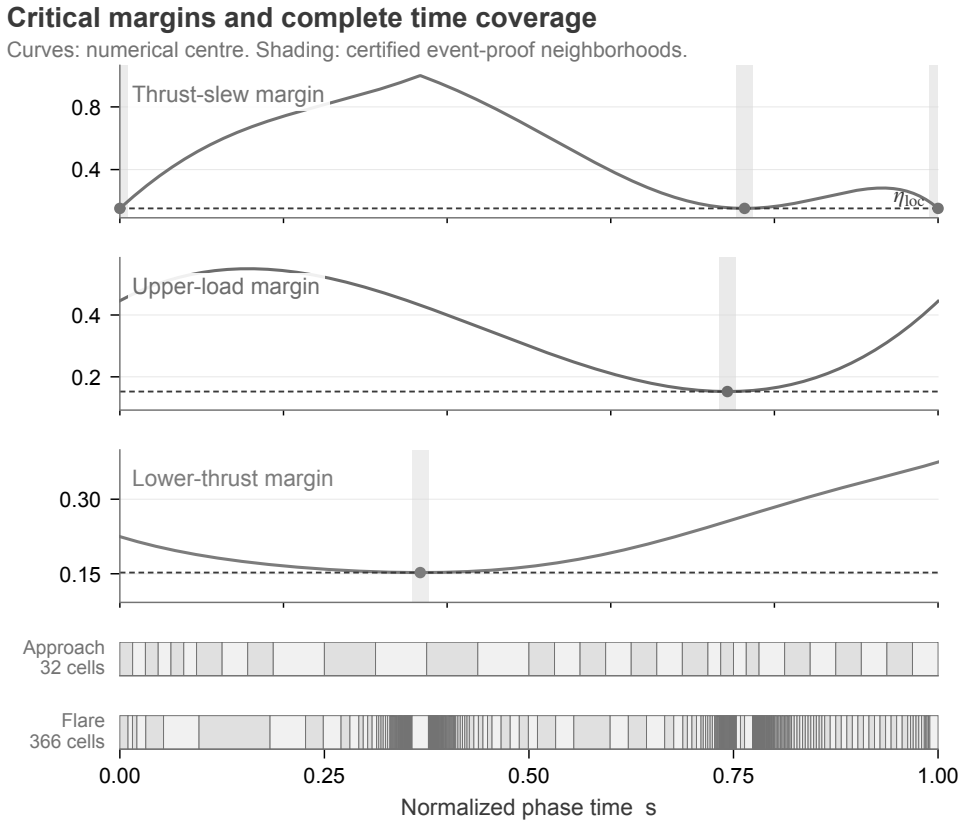


Figure 5
Critical flare margins at the numerical root centre, with the certified primary level and special event neighborhoods. Curves are plotting evaluations, not proof by sampling. The lower strip shows the accepted interval partition; analytical event bounds complete the certificate in the shaded neighborhoods.

6.3 Transfer to the complete parameter family

Theorem 4 (Non-strict local optimality in twelve parameters). The connected path $q^*=(a^*,p^*)$ of Theorem 3 is a local primary maximum in the original twelve-parameter family. More strongly, there is a neighborhood N of p^* such that every feasible (a,p) with $p\in N$ has $\eta(a,p)\le\eta_0$, with strict inequality whenever $p\neq p^*$. The conclusion is uniform over admissible approach parameters a.

Proof. The gate position, velocity and acceleration are fixed by the declared wind and endpoint conditions. The flare Hermite coefficients, its two bumps and all flare force quantities therefore depend only on p, not on any of the seven approach coordinates. The stopping margin depends only on (x_D, U). Thus the six fixed-time functions used in Theorem 2 remain necessary constraints for every complete feasible (a, p):

$$\eta(a,p) \le \psi(p) := \min_{1\le i\le 6} g_i(p).$$

Varying T_a shifts the absolute gate time but does not remove any normalized flare time from the domain. In particular the flare-side gate slew remains constrained independently of its approach-side counterpart. The strict upper-support argument proves $\psi(p) < \eta_0$ for $p \neq p^*$ sufficiently close, while $\psi(p^*) = \eta_0$. Feasibility at (a^*, p^*) supplies equality there. This proves the stated local maximum without assuming that any particular approach perturbation remains feasible. ∎

Approach-only perturbations may preserve the value, lower it, or violate feasibility. The theorem does not classify them and does not assert strict twelve-dimensional optimality. Nor does it imply local optimality with respect to arbitrary smooth flare perturbations outside the polynomial family. Combining the feasible root with Proposition 4 gives (1); the printed lower endpoint is rounded downward from the exact root-box lower endpoint.

7. Conditional results and sensitivity

7.1 Continuous primary retention

Let q_S denote the original delivered baseline secondary trajectory, retained without changing its coefficients. It is not the exact root path: its rounded touchdown and stop positions are 301.996569301 m and 2234.513819249 m.

Proposition 5 (Retention relative to the proved local primary value). The delivered secondary trajectory is continuously feasible and satisfies

$$\eta_0 - \eta(q_S) \le 10^{-4}. \tag{43}$$

This is not a proof of local or global secondary-effort minimality.

Computer-assisted proof. The root box proves $\eta_0 \le \overline{\eta} = c_6 + 10^{-20}$. The complete-time secondary checker, including hard geometry and exact ground rollout, certifies

$$\begin{aligned}\eta(q_S) &\ge \overline{\eta} - 10^{-4} \\ &= 0.152260764301861701928625206455836893973748530913.\end{aligned} \tag{44}$$

The separate calculation processes 4,020 time cells; this is a processed count, not the accepted count from the root certificate. Subtracting the certified lower bound from the certified upper bound proves (43). No comparison of sampled minima is used. The argument supplies no lower bound on the loss and no inequality comparing $\mathcal{E}(q_S)$ with other retained paths. ∎

For the other delivered primary/secondary pairs a related comparison uses an interval upper evaluation of any one primary margin at a selected time. Since a continuous minimum cannot exceed that value, its upper endpoint is a valid primary upper bound. A secondary all-time lower bound at that endpoint minus 10^{-4} proves retention relative to that particular primary path. A dense scan may select the evaluation point but cannot certify a lower bound on the continuous primary minimum. Only the baseline comparison above uses the proved exact local optimum.

7.2 Implicit sensitivity and an explicit wind family

The equations extend smoothly to $F(\zeta, w_x)$ near the baseline while w_y and every other parameter remain fixed. Theorem 1 gives nonsingularity at the root, so the implicit-function theorem supplies a local equality branch and

$$\frac{d\zeta}{dw_x} = -F_\zeta^{-1} F_{w_x}. \tag{45}$$

The saved active-branch calculation evaluates

$$\frac{d\eta}{dw_x} \simeq -0.01800313 \quad \text{per m/s}.$$

This value was computed numerically and checked by finite-difference resolves; it is not a rigorous interval enclosure of the derivative. Baseline nonsingularity alone also supplies no explicit continuation radius and no guarantee that the same active regime remains controlling throughout a displayed wind range.

A separate statement concerns an explicit feasible construction. Let q_c and q_c' be the exact decimal vectors in Appendix F, copied from the frozen numerical branch output. For $\delta = w_x - w_x^0$, set

$$q(\delta) = q_c + q_c'\delta, \qquad w(\delta) = (w_x^0 + \delta, w_y^0, 0). \tag{46}$$

The seven approach derivatives in q_c' are zero. Nevertheless the approach coefficients change with wind because its endpoint velocities are recomputed using (10). The construction is affine in the decision parameters, not in the controls, and its centre is a specified decimal vector rather than the exact implicit root.

Proposition 6 (Certified known-wind neighborhoods). For every δ in each closed interval in Table 5, the path (46) is a feasible complete A1 landing in the original parameter box, with continuous primary margin at least the corresponding lower bound.

Table 5
Certified parameter domains for the explicit family. These are feasibility levels, not bounds on the sensitivity or an optimality radius.

Along-wind perturbation (m/s)	Certified primary lower bound
$\lvert\delta\rvert \le 0.001$	0.149
$\lvert\delta\rvert \le 0.002$	0.147
$\lvert\delta\rvert \le 0.005$	0.140

Computer-assisted proof. For $j = 0,1,2,3$, interval automatic differentiation encloses $\partial_\delta R^{(j)}(s,\delta)$ over the time cell and parameter interval. The mean-value theorem gives

$$R^{(j)}(s,\delta) \in R^{(j)}(s,0) + [-d,d]\,\partial_\delta R^{(j)}(s,[-d,d]), \qquad |\delta| \le d.$$

Duration denominators and wind-dependent endpoint data are differentiated, preserving the actual parameter dependence. Subsequent interval hulls may enlarge this set but do not exclude a possible value. Each of the three computations proves complete closed time coverage, all operating margins at its stated level, hard whole-body geometry, contact factorization and exact rollout. It also verifies that every $q(\delta)$ stays inside the original parameter bounds. The supplied executable reproduction reruns these finite inclusions; it does not infer a domain from finitely many wind samples. ∎

The widest radius is about 0.00972 kt. Each wind is known and constant when constructing its associated trajectory. There is no common control law responding causally to unobserved gusts. The attempted interval enclosure at $|\delta| \le 0.01$ was unresolved; interval overestimation is not an infeasibility proof.

7.3 Ten accepted condition points

Table 6 reports the delivered secondary paths across the ten accepted conditions. Wind-only changes preserve the baseline geometry. Urban and buffer changes retain baseline wind; urban height replaces only the blanket height. The numbers are deliberately conservative downward-rounded feasibility levels, not estimates of the global optima. Every row has its own continuous flight/geometry/rollout certificate. The baseline exact root is a separate path from the baseline delivered-secondary row.

Table 6
Ten separately certified delivered paths. Margin lower bounds are rounded downward to six decimals; positions are rounded display values in metres. Wind labels denote the named real-decimal scenario vectors, with exact inputs in the package.

Condition	$\eta \geq$	x_D	x_{stop}
Baseline wind	0.152258	301.997	2234.514
Calm	0.039021	313.611	2533.008
10 kt headwind	0.146202	300.948	2250.477
2 kt tailwind	0.016768	317.644	2591.668
3 kt tailwind	0.005547	319.660	2621.245
Urban height 45 m	0.152258	301.997	2234.514
Urban height 50 m	0.152249	301.978	2234.537
Urban height 55 m	0.099436	413.404	2373.753
Urban height 58 m	0.011876	694.037	2593.470
Horizontal allowance 15 m	0.152258	301.997	2234.514

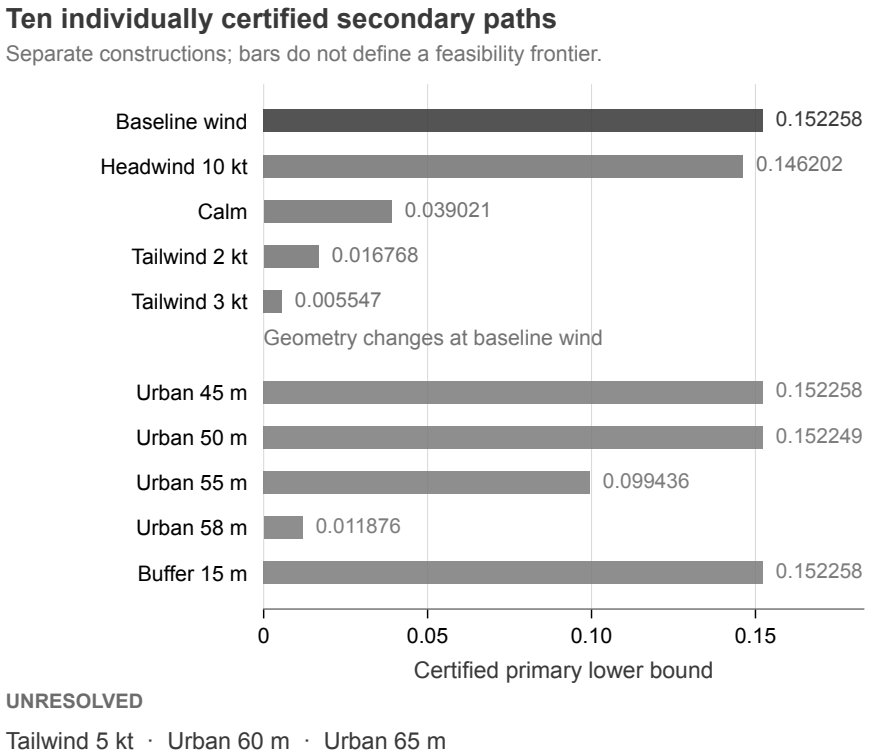


Figure 6
Individually certified condition points. Each bar represents a lower bound for a distinct delivered trajectory, not a common policy or a feasibility frontier. Tailwind 5 kt and urban heights 60 m and 65 m remain unresolved and are listed without invented numerical bounds.

A separate fixed-gate check excludes the specific frame-shift case $(+80, -50)$ m from the imposed conservative model: an urban-side predicate is -10 m at the common gate, regardless of trajectory parameters. This is an analytical endpoint obstruction, not evidence about the actual airport. By contrast, the 5 kt tailwind and the 60 m and 65 m urban-blanket cases have neither accepted feasible paths nor global exclusions. Negative dense-search margins and local condition-transition estimates do not resolve those cases.

8. Discussion and open mathematical problems

The positive result is a certified connection between an exact local optimizer and a complete feasible hybrid trajectory. Root isolation identifies an actual solution of the active equations. The gradient simplex proves strict local primary optimality in five flare coordinates. Stationary-time arguments and interval coverage establish feasibility, including the approach, occupied geometry and ideal contact law. The dependence structure then transfers the upper-support argument to a non-strict local maximum in twelve parameters.

The remaining global gap is concrete. The achieved lower endpoint in (1) and the family exclusion differ by approximately 0.0296094506981383. Closing that gap requires either a better feasible construction or a stronger exclusion using constraints discarded by the energy relaxation. The six active flare constraints suggest a lower-dimensional global search, but local sharpness does not exclude another branch. Nor does the terminal relaxation identify the controls needed to attain its scalar bound.

The secondary problem is also distinct. Continuous primary retention has been certified, and numerical effort minimizations have been delivered, but no inequality proves their effort optimality. A complete condition map would require validated continuation of active branches, treatment of changing active sets and geometric disjunctions, and either constructions or exclusions for the unresolved wind and urban cases. The three small wind neighborhoods establish only the explicitly parameterized feasible family.

The mathematical and physical domains should be kept separate. This reduced model omits calibrated aerodynamic uncertainty, six-degree-of-freedom attitude dynamics, load-rated contact, causal gust response and go-around viability. Its scene completes missing obstacle coverage through explicit assumptions. Those limitations do not negate the local theorem; they identify the object to which it applies. A historical or operational claim would require a different evidential and modelling argument.

Appendix A. Polynomial identities and regularity bounds

For a scalar coordinate the endpoint construction can be written

$$P(s) = P_0 + (P_1 - P_0)H_0(s) + Tv_0A_0(s) + Tv_1B_0(s), \tag{47}$$

where

$$H_0(s) = 10s^3 - 15s^4 + 6s^5,$$
$$A_0(s) = s - 6s^3 + 8s^4 - 3s^5,$$
$$B_0(s) = -4s^3 + 7s^4 - 3s^5.$$

Direct differentiation verifies the six endpoint conditions in (9). The bump $b = 64s^3(1-s)^3$ has a zero of order three at both endpoints; multiplying it by $2s-1$ preserves these zeros. Thus the family enforces exactly the positions, velocities and accelerations stated in the main text. No numerical linear solve is needed to establish the identities.

For reproducible rational range bounds, let $p(s) = \sum_{k=0}^{n} c_k s^k$ and substitute $s = a + ht$ on a rational cell. The power coefficients become

$$d_j = h^j \sum_{k=j}^{n} c_k \binom{k}{j} a^{k-j},$$

and the degree-n Bernstein coefficients are

$$\beta_i = \sum_{j=0}^{i} d_j \frac{\binom{i}{j}}{\binom{n}{j}}, \quad 0 \le i \le n. \tag{48}$$

Indeed expansion of the Bernstein basis gives $t^j = \sum_{i=j}^{n} \binom{i}{j}\binom{n}{j}^{-1}\binom{n}{i} t^i (1-t)^{n-i}$. Its basis functions are nonnegative and sum to one, so p lies between the smallest and largest β_i. This is a finite exact rational calculation, not a bound inferred from sampled values.

Apply (48) to H_0', A_0', B_0', b', o' on the 128 cells $[i/128, (i+1)/128]$. Maximum absolute enclosure endpoints are, respectively,

$$15/8, \quad 1, \quad 1, \quad 288141339/83886080, \quad 2.$$

The entry and gate vertical speeds have magnitude less than 5, as outward evaluation of $153k\exp(z/17000)\sin(3.1^\circ)$ at $z = 146, 21.24$ verifies. Therefore

$$|\dot{z}_a| \le \frac{124.76}{30}\frac{15}{8} + 5(1+1) + \frac{50}{30}\left(\frac{288141339}{83886080} + 2\right) = \frac{33792275107}{1258291200},$$
$$|\dot{z}_f| \le \frac{15.24}{4}\frac{15}{8} + 5 + 0.7 + \frac{10}{4}\frac{288141339}{83886080} = \frac{719106075}{33554432}.$$

The horizontal-air-speed separation can even be proved by the rational inequality

$$(135k)^2-\left(\frac{33792275107}{1258291200}\right)^2-64^2=\frac{9617823244026244151}{1583296743997440000}>0.$$

Thus $H>64$ on the nonnegative feasible set. This prevents a singular limit in the compactness proof independently of numerical rounding of (21).

For the flare height bound, $0\le H_0\le 1$, $A_0\ge 0$, $-0.2\le B_0\le 0$, and $0\le b\le 1$ on $[0,1]$; these follow either by factorization or the same rational enclosure. Since $v_{G,z}<0$, $v_{D,z}=-0.7$, $T_f\le 12$ and $f_z\le 10$,

$$z_f\le 21.24+12(0.7)(0.2)+10=32.92<34.$$

For the lower bound, $6\cos\theta+6.0579\sin\theta$ is increasing on $[0,25°]$, because its derivative is positive there. Gear nonpenetration then implies $z_f\ge 6$. These height and derivative bounds supply exactly the family-specific premises of Proposition 3.

Appendix B. Differentiating inverse dynamics

Let $a=\ddot{R}$, $j=\dddot{R}$ and $\dot{V}=e_t\cdot a$. Constant wind gives

$$\dot{e}_t=(a-e_t\dot{V})/V, \qquad \dot{\psi}=(v_x a_y-v_y a_x)/H^2,$$
$$\dot{e}_r=-\dot{\psi}(v_x,v_y,0)/H, \qquad \dot{e}_u=\dot{e}_t\times e_r+e_t\times\dot{e}_r.$$

With $\dot{F}=mj$,

$$\dot{L}_u=\dot{F}\cdot e_u+F\cdot\dot{e}_u, \qquad \dot{L}_r=\dot{F}\cdot e_r+F\cdot\dot{e}_r,$$
$$\dot{L}=(L_u\dot{L}_u+L_r\dot{L}_r)/L, \qquad \dot{Q}=Q(-\dot{z}/8500+2\dot{V}/V),$$
$$\dot{C}_L=\dot{L}/Q-C_L\dot{Q}/Q,$$
$$\dot{D}=\dot{Q}(0.08+0.045C_L^2)+0.09QC_L\dot{C}_L,$$
$$\dot{T}=\dot{D}+\dot{F}\cdot e_t+F\cdot\dot{e}_t.$$

Together with (8), these are the exact first-rate formulas used by the continuous checker and the independent floating-point evaluator. They hold on each open phase and extend one-sided to its endpoints. Smooth position through acceleration does not imply a shared jerk, so omitting either side of a rate check at the gate would change the model.

In the flare, $a_y=0$ and $v_y=-w_y$. Thus $L_r=mw_y a_x/H$, the identity used in (42). Zero endpoint accelerations and $a_y=0$ imply zero bank at touchdown even with crosswind: air heading and ground heading need not coincide, but the lateral projection of the vertical weight-support force onto e_r is zero.

Appendix C. Exact root and fixed approach data

The decimal strings in Table A1 define c exactly; they are not rounded coefficients of an assumed exact optimizer. The root is the unique solution of (32) in (34). Its coordinates are correlated by those equations. An independent selection of interval endpoints generally does not solve them. In the same

Table A1
Exact decimal centre of the scaled root box. Every coordinate has exact radius 10^{-20}.

Coordinate	Exact centre c_i
$T_f/5$	0.828166892572350798346132239036969796434284587227
$x_D/300$	1.007088197975782777579889230796463121574943185 37
$U/75$	0.966828564224417027917923697606192868395258198399
f_x	-0.054163922081677332101141370755837503506443770627 7
f_z	-0.014824733205613137679936250554683650603549289457 3
η	0.152360764301861701918625206455836893973748530913
s_S	0.763520832037558933983831253901580196263121832139
s_L	0.742524712666685897431892059100460080408671780144
s_T	0.367159602316114440267379742208002318715902381307

coordinate order, outward bounds on $|K_i - c_i|$ are 10^{-36} times

$$(6.762,\ 7.685,\ 1.594,\ 15.48,\ 196.4,$$

$$1.696,\ 116.2,\ 33.95,\ 53.98).$$

The seven approach coordinates, in the units of Table 1, are the following exact finite decimals from the baseline active-branch input:

$$\begin{aligned} T_a &= 40.45511499029442 \\ b_x &= 17.554312278350075 \\ c_x &= -18.902355338597157 \\ b_y &= 24.330223470703103 \\ c_y &= 6.9036241051397464 \\ b_z &= 45.306069713015454 \\ c_z &= -1.453757342378532 \end{aligned}$$

They are held fixed only to construct one connected feasible path. Theorem 4 does not require all nearby approaches to remain feasible.

For avoidance of decimal ambiguity, the full primary box endpoints are

$$\underline{\eta} = 0.152360764301861701908625206455836893973748530913,$$

$$\overline{\eta} = 0.152360764301861701928625206455836893973748530913.$$

The physical root-coordinate radii inherited from the scaled box are 5×10^{-20} s for T_f, 3×10^{-18} m for x_D, 7.5×10^{-19} m/s for U, and 10^{-20} m for each flare bump. These are isolation widths, not physical uncertainty or a local-optimality radius. The much smaller Krawczyk image demonstrates strict inclusion but is not needed as a new advertised parameter box.

Appendix D. Interval automatic differentiation and Taylor jets

The interval evaluator uses two nested exact algebraic constructions. A dual number stores an interval value a and nine derivative intervals a_i. Addition and multiplication obey

$$(ab)_i = a_i b + a b_i, \qquad (a/b)_i = (a_i b - a b_i)/b^2,$$

with the usual derivative formulas for square root and exponential on their regular domains. Each operation is evaluated with outward rounding. Induction on the expression graph proves inclusion of the value and each derivative.

The normalized-time jet stores coefficients $a_n = a^{(n)}(s)/n!$, $0 \le n \le 3$, each itself a dual number. If $c = ab$, $b = 1/a$, $b = \sqrt{a}$ or $b = \exp a$, the coefficient recurrences are

$$c_n = \sum_{k=0}^{n} a_k b_{n-k},$$

$$b_0 = 1/a_0, \qquad b_n = -a_0^{-1} \sum_{k=1}^{n} a_k b_{n-k} \qquad (b = 1/a),$$

$$b_0 = \sqrt{a_0}, \qquad b_n = \frac{a_n - \sum_{k=1}^{n-1} b_k b_{n-k}}{2 b_0} \qquad (b = \sqrt{a}),$$

$$b_0 = \exp(a_0), \qquad b_n = \frac{1}{n} \sum_{k=1}^{n} k a_k b_{n-k} \qquad (b = \exp a).$$

The product identity, $ab = 1$, $b^2 = a$ and $b' = a'b$ prove these recurrences coefficient by coefficient. Keeping order three loses no derivative of order at most three. Position polynomials themselves are evaluated without approximation; the jet truncation is a derivative representation, not a truncated trajectory.

Applied to thrust, the jet gives T_s, T_{ss}, T_{sss}. Its dual components supply the mixed derivatives for DF, including derivatives of the stationarity equations with respect to all root coordinates. The interior slew curvature is $-T_{sss}/(100000 T_f)$ on the positive branch. Load and lower-thrust curvatures use their second derivatives. This is why finite-difference Jacobians are unnecessary in the proof.

The preconditioner is chosen numerically for sharpness, but its final finite decimal entries are exact specified constants. The inequality $\|I - C\mathbb{J}\|_\infty < 1$ is checked outward, and therefore also proves C nonsingular; no unverified inverse is assumed. The additive matrix export in the package records every interval Jacobian entry, both decimal preconditioners, residual matrices and exact binary endpoint representations. Reproduction recomputes them from the source expressions.

Appendix E. Continuous-time verification algorithm

The verifier has two logically separate inputs: exact polynomial/physical data and a finite list of proved active neighborhoods. Its procedure is:

1. Enclose the root parameters and construct interval Hermite coefficients, with exact endpoint identities retained symbolically.
2. Split $[0,1]$ at every special-event neighborhood endpoint. Treat approach and flare as separate closed domains.
3. On each cell, enclose position through jerk, inverse forces, exact rates, corridor distance and all geometric predicates. Remove only a margin whose entire cell lies in its proved neighborhood.
4. Accept only after all remaining margin lower endpoints exceed $\overline{\eta}$, and all hard geometry lower endpoints are nonnegative. Otherwise bisect, or return unresolved at the depth limit.
5. Check that the accepted closed cells cover both phases without gaps; check terminal margins, factored gear contact and the whole ground segment.

The maximum accepted subdivision depth in the local certificate is 12. Cells are enclosures of all real times in their domains, not quadrature nodes. The exact numerical value of a selected subdivision endpoint is its stored decimal representation, consistently used in both event and ordinary interval evaluations.

The disjunctive geometry extension encloses maxima and minima by taking the corresponding extrema of endpoint bounds. It is inclusion-preserving but can be pessimistic. The family dependence is likewise enclosed, not discarded: an interval hull may include combinations no actual parameter realizes, which can make certification harder but cannot make a false trajectory feasible. For the wind-family check, first-order parameter differentiation reduces that dependency inflation before the force enclosure.

Appendix F. Explicit wind-family coefficients

The following finite decimals, not the exact local root, define (46). Nonzero derivative entries have the units of the corresponding parameter divided by m/s. Entry and gate velocities are recomputed from the changed known wind.

Table A2
Exact finite-decimal wind-family input vectors.

Coordinate	Exact q_c decimal	Exact q'_c decimal
T_a	40.45511499029442	0
T_f	4.1408344628617435	-0.11066452444996222
x_D	302.1264593927342	-3.935299293324053
U	72.51214231683133	1.0018913574888975
b_x	17.554312278350075	0
c_x	-18.902355338597157	0
b_y	24.330223470703103	0
c_y	6.9036241051397464	0
b_z	45.306069713015454	0
c_z	-1.453757342378532	0
f_x	-0.05416392208167613	0.018454391224874828
f_z	-0.014824733205644995	-0.22842502574710072

The baseline root-isolation radius 10^{-20}, the unquantified local-optimality neighborhood and the wind intervals in Table 5 are three different domains. None can be substituted for either of the others.

Appendix G. Source chronology, invalidations and claim domains

The source freeze is public repository commit `78a9ce1c627f917c94306af5529d5e9ebeb743e8`. The latest mathematical layer at that commit is the A1 local certificate, not the older analytical status in V5's frozen evidence. The note preserves that chronology.

The original synthetic model had a continuously certified witness with margin at least 0.0397 over 2,502 closed cells and a family bound 0.1559772, with a broader terminal-only bound near 0.15769490255. These values use its own altitude convention and zero-wind scenario. They are not competing bounds on A1 at the baseline wind. The A1 terminal floor value (20), its broader terminal relaxation near 0.223501289399, the earlier energy upper bound 0.191605 and the later 0.181970215 each have their stated domains.

Two earlier global-checker failures are retained as invalidated evidence. A corridor-distance lower enclosure was incorrectly used for an exclusion upper bound, and interval `atan2` across its branch cut wrongly excluded a box containing a known feasible witness. The associated claimed upper values 0.10, 0.05, 0.04 and 0.03972 were withdrawn. They are not used here. The repaired checker uses a conservative full-angle hull when the denominator interval is nonpositive; the accepted candidate additionally has strictly positive lift-up and horizontal speed.

An earlier urban55 search missed the one-sided clearance just before exiting the upstream urban region. Its pre-repair candidate is preserved. The accepted path has an explicit exit-root condition and a fresh continuous certificate. Failed wider wind enclosures and negative-margin tailwind/urban searches are also preserved, without being promoted to infeasibility results. These records

explain why accepted input hashes and finite proof coverage matter more than an optimizer success flag.

Appendix H. Certificate specification and reproduction

The source package includes the manuscript, original vector figure code, compact figure inputs, V5's unchanged academic class/style files and a manifest-bound public proof-source subset. No private source scans or installable proprietary fonts are included. Relative paths below are rooted in the package. The source subset preserves repository-relative paths beneath `frozen-source/`.

Object	Role and finite content
`inputs/local_certificate.json`	Exact decimal centre/radius, Krawczyk offsets, multiplier intervals, event inequalities and 398-cell cover.
`verification/accepted/`	Fresh reproduction receipts and logs; full matrices are in `verification/final/local_matrices.json`.
`frozen-source/PUBLIC_SOURCE_MANIFEST.json`	SHA-256 binding of every included public source file to the pinned commit.
`CLAIMS_AND_EVIDENCE.csv`	Claim classification, premises, executable evidence and theorem linkage.
`audit/`	Independent analytical/model, interval/optimization and numerical-reproduction reviews; corrections and failures retained.
`FIGURE_PROVENANCE.md`	Per-figure mathematical object, input, plotting status and intended interpretation.

The build and proof checks are separate. After creating an environment from the supplied pinned requirements, the principal commands are

```
python build.py --figures
python scripts/verify_all.py \
  --source-root frozen-source \
  --output-dir verification/replay
python verification/emit_matrices.py \
  --source-root frozen-source \
  --output verification/replay/local_matrices.json
```

The README gives exact invocation details, host-tool prerequisites and the tested environment. The build uses XeLaTeX, BibTeX, locally licensed Times/Arial/Courier New and NewTX mathematics. The class retains the V5 page trim 182.52×274.92 mm, 373.2 TeX-point text width, 12-point body and 14.4-point normal baseline. Source package extraction is checked in a fresh directory; the existing host TeX and licensed fonts are reused, not claimed to be a fresh operating-system installation.

The publicly reproducible numerical suite contains 102 numerical/core/proof tests and 15 geometry tests. Five historical-source tests and two original-workspace preservation tests require omitted or unnormalized inputs;

they are explicitly excluded, not counted as passes. Publication mapping and the new source manifest supply separate integrity checks. The old publication checker expects a historical complete project ledger; later V5 appendages make that exact-file expectation stale. The additive verification checks the old prefix and the pinned current file separately, preserving the original failure log and checker.

Root, simplex, event and complete-path calculations are freshly executed, as are baseline secondary retention, the stronger global exclusion, the three wind domains and the ten accepted condition paths. Restricted historical originals are not needed for those conditional mathematical statements.

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